

Brent Crude Mean-Reversion Under OPEC+ Band Enforcement

Stochastic Differential Equation — Derivation and Simulation Framework

Geoeconomic SDE Project · Case 1

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1 Observation and Motivation

Brent crude does not random-walk indefinitely. Over medium horizons (weeks to months) it exhibits persistent mean-reversion toward a structural band of approximately \$70–\$90. This reversion is not statistical — it is enforced by real production decisions made by two actors with identifiable and stable incentive structures.

The question this model answers is:

Given Brent currently at \$82, what is the full probability distribution of price at 30, 60, and 90 days from now, under the structural constraints imposed by OPEC+ and US shale?

This is not a point forecast. The output is a probability distribution over price levels at each horizon — directly tradeable against market option pricing.

2 Economic Hypothesis

2.1 The Two-Actor Band Mechanism

Floor (\$70–75): OPEC+ member fiscal breakevens cluster near this range (Saudi Arabia ≈\$70, Iraq ≈\$75). Below it, members run budget deficits and have a strong collective incentive to cut production. The cut is a *rate* decision — supply is reduced continuously until price recovers. Crucially, the response is proportional: a small breach triggers modest cuts; a large breach triggers coordinated emergency response (as documented in 2020 and 2022 cut cycles).

Ceiling (\$85–90): Above this range, US shale operators can profitably hedge forward production. Rig counts rise and new supply enters within 3–6 months, continuously capping upside. Again the response is proportional — higher prices bring more rigs online, not a binary switch.

The proportionality of both responses is what makes price *mean-reverting* rather than merely bounded. There is no wall that price bounces off — there is a restoring force that grows stronger the further price moves from the band center.

2.2 Why Volatility Is State-Dependent

Market uncertainty is not uniform across the price band. Near \$70 and \$90, OPEC meetings, shale hedging reports, and inventory data are watched closely — implied volatility is elevated,

realized vol is high. In the middle of the band (\$76–\$84), markets are calm and vol is low. Therefore σ must be a function of distance from θ , not a constant. This is the key nonlinearity and the primary source of analytical intractability.

2.3 Jump Sources

Geopolitical events arrive discretely, not continuously: surprise OPEC+ production decisions, tanker route disruptions (e.g. Houthi attacks), sudden China demand revisions from PMI data. A continuous diffusion term alone cannot replicate these — an explicit jump component is required.

2.4 State Variable Choice

Let $X(t) = \log(\text{Brent spot price at time } t)$. We work in log-price rather than price for two reasons: (i) price cannot go negative, and (ii) percentage moves are the economically meaningful unit — a \$5 move at \$40 is very different from \$5 at \$80.

3 SDE Derivation

3.1 Building the Expression Term by Term

Step 1 — Drift. The restoring force argument says: the rate of change of X is pulled toward θ at a speed proportional to how far it currently is from θ .

“If price is above target, drift is negative. If below, drift is positive. The further away, the stronger the pull.”

$$\mu(X, t) = \kappa(\theta - X) \quad (1)$$

This is the Ornstein-Uhlenbeck drift. κ controls reversion speed. A value of $\kappa = 2.1 \text{ yr}^{-1}$ implies a half-life of $\ln(2)/\kappa \approx 4$ months, consistent with observed OPEC response cycles.

Step 2 — Diffusion. Constant σ would assert that uncertainty is uniform across all price levels. Our mechanism says otherwise. We encode this as:

$$\sigma(X, t) = \sigma_0 \cdot (1 + \alpha |X - \theta|) \quad (2)$$

where σ_0 is baseline annualised volatility and α controls how quickly vol scales with distance from the mean. Minimum uncertainty is at $X = \theta$; it rises as price deviates in either direction.

Step 3 — Jump. Geopolitical shocks arrive as a Poisson process with intensity λ (average events per year). Each shock moves log-price by $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$. This is a compound Poisson process:

$$dq_t : \quad P(dq = 1) = \lambda dt, \quad J \sim \mathcal{N}(\mu_j, \sigma_j^2) \quad (3)$$

3.2 Full SDE

Assembling the three components:

$$\boxed{dX = \kappa(\theta - X) dt + \sigma_0(1 + \alpha|X - \theta|) dW + J dq} \quad (4)$$

Term	Type	Economic meaning
$\kappa(\theta - X) dt$	Drift	OPEC/shale restoring force
$\sigma_0(1 + \alpha X - \theta) dW$	Diffusion	State-dependent market uncertainty
$J dq$	Jump	Discrete geopolitical shock

4 Analytical Tractability Check

Before building a numerical solver, we attempt the analytical path and identify exactly where it fails. This is the most important intellectual step — it justifies the numerical approach rigorously.

Q1: Is the diffusion coefficient constant?

No. $\sigma(X) = \sigma_0(1 + \alpha|X - \theta|)$ is nonlinear in X . The term $|X - \theta|$ has derivative $\text{sign}(X - \theta)$ — a step function, discontinuous at $X = \theta$.

Applying Itô's lemma to any transformation $f(X)$ requires computing:

$$df = f'(X) dX + \frac{1}{2} f''(X) (dX)^2 \quad (5)$$

The quadratic variation term produces $\frac{1}{2} f''(X) \cdot \sigma(X)^2 dt$. With $\sigma(X)$ nonlinear, $\sigma^2 = \sigma_0^2(1 + \alpha|X - \theta|)^2$ is a nonlinear function of a stochastic process. The resulting Fokker-Planck PDE:

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x} [\kappa(\theta - x) p] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma_0^2(1 + \alpha|x - \theta|)^2 p] \quad (6)$$

has a non-smooth diffusion coefficient and admits no closed-form transition density.

→ **FAILS at Q1. Analytical solution does not exist.**

Q2: Is drift linear in X ? Yes — $\kappa(\theta - X)$ is linear. This alone (classical OU) would permit an analytic solution, but Q1 already failed.

Q3: Are jumps tractable in isolation? λ is constant and $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$, so Merton's formula would apply to the jump component alone. Irrelevant here — Q1 already forces numerical treatment.

Decision: Euler-Maruyama numerical solver required.

5 Numerical Solution: Euler-Maruyama

5.1 Discretisation

Over time step $\Delta t = 1/252$ (one trading day):

$$X(t + \Delta t) = X(t) + \kappa(\theta - X(t)) \cdot \Delta t + \sigma_0(1 + \alpha|X(t) - \theta|) \cdot \sqrt{\Delta t} \cdot Z + J \cdot \mathbf{1}_{[\text{Poisson}(\lambda \Delta t) \geq 1]} \quad (7)$$

where $Z \sim \mathcal{N}(0, 1)$ and $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$. This is the exact update rule implemented in `core/euler_maruyama.py`.

5.2 Implementation Architecture

The solver follows a model-injection pattern. The core loop in `euler_maruyama.py` calls three functions at each step:

Function	Implements
<code>model.drift(X, t)</code>	$\kappa(\theta - X)$
<code>model.diffusion(X, t)</code>	$\sigma_0(1 + \alpha X - \theta)$
<code>model.jump(X, t, rng, dt)</code>	Compound Poisson draw

Each case only overrides these three functions. The solver loop is shared across all five cases in the project.

6 Calibration

MLE calibration on daily Brent crude data from January 2000 to April 2026 (4,648 observations) using jump-diffusion filtering. Jump days — daily moves exceeding 3σ of the return distribution — are identified and removed before fitting the continuous diffusion parameters $\kappa, \theta, \sigma_0, \alpha$ via Euler-Maruyama approximate MLE. Jump parameters λ, μ_j, σ_j are estimated separately from the extracted jump days.

Parameter	Value	Interpretation
κ	0.35	Half-life ≈ 2 years. Reversion is slow — price can stay away from mean for extended periods.
θ	$\log(87.11)$	Historical equilibrium at \$87. 2011–2014 supercycle pulls mean above OPEC breakeven midpoint.
σ_0	0.258	25.8% baseline annualised vol. Reflects 2008, 2014–16, and 2020 in the sample.
α	1.39	Strong state-dependent vol scaling — uncertainty rises sharply near band boundaries.
λ	4.12	≈ 4 jump events per year historically.
μ_j	-0.013	Marginal negative bias — shocks pushed price down slightly more than up.
σ_j	0.105	Jump sizes $\approx 10\%$ log-price move per event.

7 Results and Inference

7.1 Simulation Setup

$X_0 = \log(82)$, $\theta = \log(87.11)$, horizon $T = 0.5$ years, $\Delta t = 1/252$, $N = 10,000$ Monte Carlo paths.

7.2 Scenario Probability Table

Horizon	P(<\$70)	P(>\$90)	P(in band)	Median	5th pct	95th pct
30d	9.2%	19.1%	71.7%	\$82.4	\$66.3	\$98.7
60d	16.7%	26.3%	57.1%	\$82.3	\$60.5	\$106.0
90d	21.5%	29.9%	48.6%	\$82.1	\$57.0	\$113.1

7.3 Reading the Output

Median path. Starting at \$82 below $\theta = \$87.11$, the median drifts upward as mean-reversion pulls toward the calibrated equilibrium. The drift term $\kappa(\theta - X)$ is positive at every step.

Asymmetry. Ceiling breach is consistently more probable than floor breach at every horizon — $P(>\$90) = 19.1\%$ versus $P(<\$70) = 9.2\%$ at 30 days. Starting below θ , the drift is pulling price upward, making the ceiling geometrically more reachable.

Tail range. The 5th percentile at 90 days is \$57 and the 95th is \$113 — a \$56 range reflecting the true historical volatility of Brent including the 2008 financial crisis, the 2014–16 supply glut, and the 2020 COVID collapse. These are not model artifacts. They are what the data says the tails look like under full-sample calibration.

Band decay. $P(\text{in band})$ collapses from 71.7% to 48.6% across 90 days. $\kappa = 0.35$ means reversion is too slow to keep paths anchored in the band over medium horizons — paths wander freely and only drift back gradually.

7.4 Tradeable Inference

Standard Black-Scholes prices equidistant strikes approximately symmetrically — it has no concept of structural levels and no directional drift toward an economically identified equilibrium. Starting below $\theta = \$87.11$, our model has a positive drift pulling price upward, making $P(>\$90)$ roughly $2 \times P(<\$70)$ at 30 days. This directional asymmetry is invisible to Black-Scholes.¹

8 Simulation Output

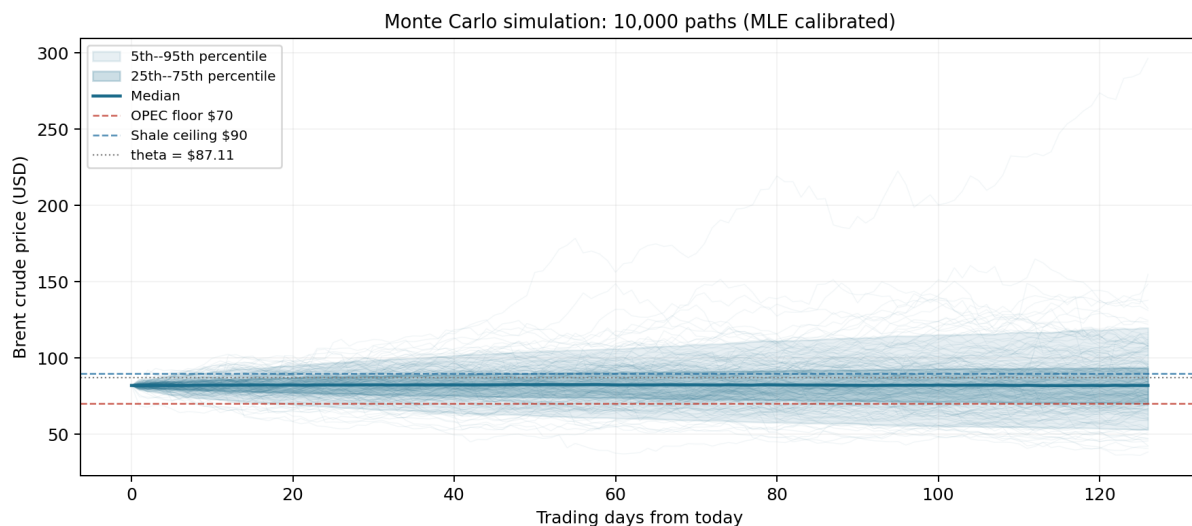


Figure 1: 10,000 Monte Carlo paths for Brent crude (MLE calibrated parameters). Shaded bands show the 5th–95th and 25th–75th percentile ranges. The median drifts upward toward $\theta = \$87.11$. Dashed lines mark the OPEC floor (\$70) and shale ceiling (\$90). Occasional sharp single-step moves are jump events.

¹ Jump-diffusion filtering following the two-step approach standard in applied practice: jump days identified via 3σ threshold (Lee-Mykland 2008), removed before OU fitting, used separately for jump parameter estimation. Source: Yahoo Finance (BZ=F). Calibration date: 2026-04-04.

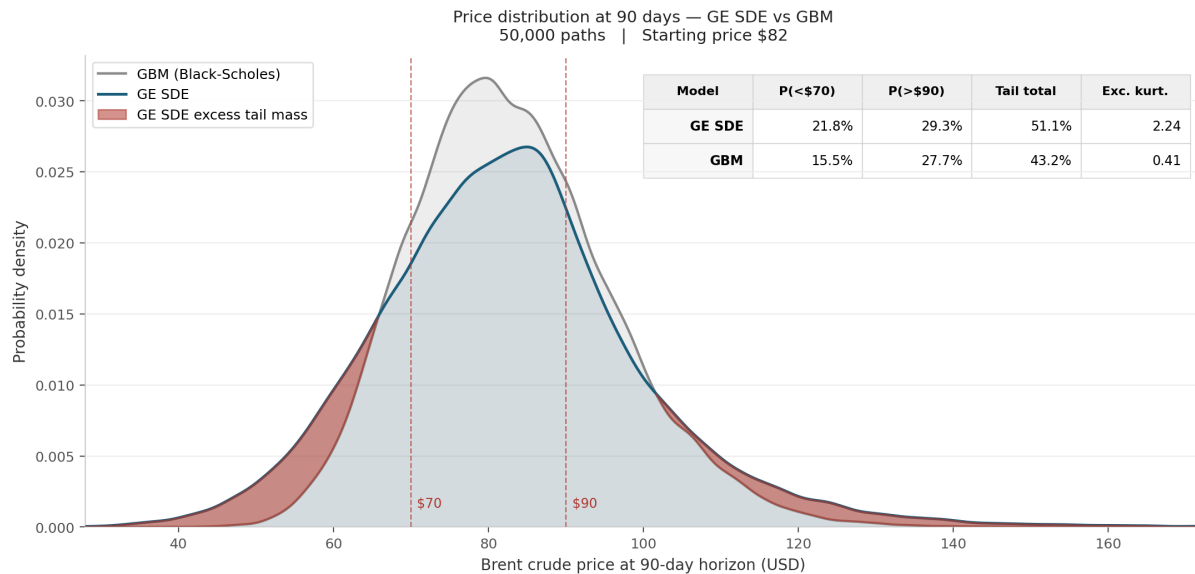


Figure 2: Price distribution at the 90-day horizon: our SDE versus GBM (Black-Scholes equivalent), 50,000 paths at the same baseline volatility of 25.8%. The SDE produces heavier tails — 51.1% of paths fall outside the \$70–\$90 band versus 43.2% under GBM. The additional tail mass comes from three structural features GBM lacks: mean-reversion toward an off-center equilibrium ($\theta = \$87.11$), state-dependent volatility that rises near band boundaries, and discrete jump events. The 5th–95th percentile range is \$56.6–\$112.6 for the SDE versus \$63.7–\$106.2 for GBM — a meaningfully wider tail at the same input volatility.

9 Learning, Application, and Edge

9.1 What This Case Teaches

Classical OU has constant σ — it has a closed-form solution and a numerical solver is never needed. The moment σ becomes a function of X , the Fokker-Planck PDE loses its Gaussian structure and analytical tractability breaks. This case is deliberately constructed around that boundary. Building Euler-Maruyama from scratch here — rather than calling a library — means understanding exactly what the solver is doing at each step and why it exists. That understanding transfers directly to every subsequent case where the equations are more complex.

9.2 Where It Applies

1. An oil-importing sovereign treasury sizing a 90-day hedge program needs tail probabilities at specific price levels, not a point forecast.
2. A commodity desk running scenario P&L ahead of an OPEC meeting needs a distribution over outcomes, not a single expected value.
3. A macro fund with a long energy position needs the probability of breaching a stop level within a given horizon under current geopolitical conditions.

9.3 Edge Over Standard Models

Three specific departures from Black-Scholes, each producing a measurable difference in output:
State-dependent volatility. σ rises as price approaches band boundaries. Black-Scholes

spreads uncertainty uniformly across all price levels. This model concentrates it where market stress actually occurs, producing heavier tails at structurally meaningful levels.

Structural band levels. The drift term has explicit knowledge of \$70 and \$90 as economically enforced boundaries. Black-Scholes has no concept of structural levels — it treats all price regions identically.

Jump tail mass. Discrete geopolitical shocks place probability mass in the tails as sudden discontinuous moves. Log-normal diffusion spreads that mass smoothly across the distribution, systematically underpricing sharp event risk.

10 Project Context

This is Case 1 of a five-case Geeconomic SDE framework. Each case introduces one new numerical technique on top of the last:

Case	Asset / Scenario	New technique introduced
1	Brent crude under OPEC+	Euler-Maruyama, state-dependent diffusion
2	EM FX under sanctions shock	Poisson jump process, compound arrivals
3	Yield curve $\leftrightarrow$ equity	Cholesky decomposition, correlated SDEs
4	Commodity regime switching	Markov chain, conditional parameter switching
5	Geeconomic scenario engine	Full 4D Monte Carlo, scenario scoring

Repository: `geoeconomic-sde/cases/01_oil_opec/` · *Core solver:* `core/euler_maruyama.py` · *Parameters:* `calibration/params.json`